

Big Data Final Evaluation Project

Students will work in teams to complete a data analysis project based on a real-world dataset. The project should demonstrate the ability to use a full big data stack including:

- **HDFS** for data storage
- **PySpark** for data processing and analysis
- **Hive** for data warehousing and querying (optional)
- **Spark SQL** for structured data operations
- **Python visualization libraries** (e.g., matplotlib, seaborn, plotly) to present insights

The goal is to show mastery of collecting, cleaning, transforming, analyzing, and communicating insights from large-scale datasets in a distributed computing environment.

Group Pairing

Submit your team members' names by the specified deadline in the WeChat group. Each team may have a maximum of two students.

Topic



Please select one of the following four project topics for your final assignment. Each group should choose only one topic to work on.

Food Analysis

Size: ~500 MB ($\approx$ 300,000+ food products, 180+ countries)

<https://www.kaggle.com/datasets/openfoodfacts/world-food-facts>

Steam Review Analysis

Size: ~50 MB ($\approx$ 27,000 games, reviews, tags, prices)

<http://kaggle.com/datasets/nikdavis/steam-store-games/data>

Book Analysis

Size: ~25 MB ($\approx$ 10,000 books, 6 million ratings)

<https://www.kaggle.com/datasets/zygmunt/goodbooks-10k>

Amazon Review Data

<https://nijianmo.github.io/amazon/index.html>

Size: ~50 GB full; *choose category subset (Books, Electronics, Beauty, etc.) $\approx$ 500 MB–2 GB*

Submission Requirements

1. Project Report (PDF)

- Introduction to your dataset
- Data storage and access strategy using HDFS
- Data preprocessing steps (cleaning, parsing, missing values)
- Analytical questions and methods (Spark, Hive, SQL, etc.)
- Key findings with explanations
- Visualizations and interpretation
- Conclusion and reflection

2. Source Code

- `.ipynb` notebook or `.py` scripts
- SQL or HiveQL scripts (if applicable)
- Data loading and cleaning pipeline using Spark/HDFS

3. Bonus (Optional)

- Dashboard or web interface
- Accessible website URL
- Your team name in the website to identify your group

Final Project Grading Rubric (Total: 100 points)

1. Project Report (50 points)

- Clear introduction of the dataset (5)
- Appropriate strategy for data storage and access using HDFS (5)

- Complete data preprocessing steps, including cleaning, parsing, and handling missing values (8)
- Well-designed analytical questions and correct use of methods (Spark, Hive, SQL, etc.) (8)
- Key findings explained with clarity and depth (6)
- Effective visualizations with meaningful interpretation (8)
- Conclusion and reflection with critical insight (5)
- Overall report coherence, clear explanation of team roles, and quality of presentation (5)

2. **Source Code** (50 points)

- Complete and executable notebook or scripts (.ipynb or .py) (10)
- Proper design of the data loading and cleaning pipeline using Spark/HDFS (10)
- Correct and effective use of Spark/HDFS tools (10)
- Code quality and style, including structure and comments (5)
- SQL or HiveQL scripts, where applicable (5)
- Evidence of methodological innovation or depth in approach (5)
- Completeness of results and effectiveness of demonstration (5)
- Optional extended implementation (e.g., dashboard or web interface) may earn up to 5 extra points, counted under this section.